Bayesian Neural Surrogates Unify Uncertainty Propagation, Sensitivity Attribution and Calibration for Coupled Multi-Physics Reactor Analysis

贝叶斯神经代理统一耦合多物理反应堆分析中的
不确定性传播、敏感性归因与标定

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Abstract / 摘要

We train a Bayesian neural network (BNN) surrogate on coupled OpenMC–FEniCS data of a heat-pipe-cooled microreactor and use its posterior predictive distribution as the single probabilistic object on which forward uncertainty propagation, Sobol sensitivity attribution and MCMC posterior calibration all operate, without retraining or recalibration between analyses. Two-way neutronic–thermal–structural coupling damps stress variability by approximately 28% relative to a decoupled prediction; stress and reactivity are governed by largely separate material parameters ($E_{\text{intercept}}$ and α_{base}, respectively); and across 18 benchmark calibrations the posterior contracts toward observation-compatible regions with 90%-CI coverage of 0.917. Each posterior predictive draw costs approximately 1.76×10^5 times less than a coupled high-fidelity solve, enabling the full uncertainty workflow to run end-to-end on a single workstation. The posterior-predictive framework is directly applicable to other coupled multi-physics systems.

本文在耦合 OpenMC–FEniCS 热管冷却微型反应堆数据上训练贝叶斯神经网络（BNN）代理，以其后验预测分布作为前向不确定性传播、Sobol 敏感性归因与 MCMC 后验标定三类分析共用的概率对象，三者之间无需各自重训或重新校准。双向中子–热–力耦合使应力变异相对解耦预测降低约 28%；应力与反应性分别由杨氏模量截距（$E_{\text{intercept}}$）与热膨胀系数（α_{base}）主导；在 18 例基准标定中，后验分布收敛至与观测相容的区域，90%-CI 覆盖率为 0.917。每次后验预测抽样的代价较耦合高保真求解降低约 1.76×10^5 倍，完整的不确定性工作流可在单台工作站上端到端运行。该后验预测框架可直接推广至其他耦合多物理系统。

1 Introduction / 引言

Microreactors—compact, transportable fission systems rated at 1–20 MW_e—are being developed for decentralized power generation in remote communities, defense installations and space missions[1,11]. NASA and DOE are advancing fission surface-power systems for sustained lunar and planetary operations[12], while rapidly growing electricity demand from AI-intensive data centres reinforces the case for distributed, dispatchable zero-carbon generation[13]. Unlike conventional large-scale plants, these designs concentrate fission power, heat removal and structural loading within a compact monolithic core where neutronics, heat transfer and structural mechanics are tightly coupled through two-way feedback[1,2]. A single uncertain material parameter can simultaneously alter the neutronic balance, the thermal field and the stress state; the closed feedback loop may amplify or damp the resulting variability in ways that single-physics analyses cannot capture. The uncertainty problem in these systems is therefore inherently coupled.

微型反应堆——额定 1–20 MW_e 的紧凑、可运输裂变系统——正面向偏远社区、国防设施及太空任务中的分布式供能而开发[1,11]。NASA 与 DOE 正推进月面及行星任务的裂变地表供电系统[12]，与此同时 AI 密集型数据中心电力需求的快速增长进一步强化了分布式、可调度零碳发电的需求[13]。与常规大型电厂不同，此类设计将裂变功率、热量排出与结构载荷集中于紧凑的一体式堆芯，中子学、传热与结构力学通过双向反馈紧密耦合[1,2]。单个不确定材料参数可同时改变中子平衡、温度场与应力状态；闭合反馈回路可能以单物理分析无法捕捉的方式放大或衰减由此产生的变异。因此，此类系统的不确定性问题本质上是耦合的。

High-fidelity coupled solvers now exist for these systems: MOOSE-based toolchains handle neutronics–thermo-mechanics–heat-pipe coupling for KRUSTY-class and larger designs[2,3], and thermal–structural analyses have established stress concentration as a persistent safety concern in monolithic-core configurations[4,5]. Surrogate-assisted probabilistic modules in the MOOSE ecosystem[6,7] and digital-twin pipelines[8] address forward propagation, sensitivity or calibration individually, but each uses a separately constructed model with no shared probabilistic basis between analyses.

高保真耦合求解器已适用于这类系统：基于 MOOSE 的工具链可处理 KRUSTY 类及更大设计上的中子–热力–热管耦合[2,3]；热–力分析已确认应力集中是一体式堆芯构型中持续性的安全关注[4,5]。MOOSE 生态[6,7] 与数字孪生管线[8] 中的代理辅助概率模块分别处理前向传播、敏感性或标定，但每类分析各自构建独立模型，三者之间缺乏共享的概率基础。

Running the full probabilistic workflow—forward uncertainty propagation, global sensitivity analysis and Bayesian calibration—directly on the coupled solver is impractical at $\sim 10^3$ s per solve. Gaussian-process and polynomial-chaos surrogates face dimensionality constraints in nonlinear coupled settings; physics-informed neural networks require a closed residual form that iterative solver exchange does not supply; deterministic neural networks produce only point predictions. None of these provides the coherent predictive distribution that all three analyses require.

在耦合求解器上直接运行完整的概率工作流——前向不确定性传播、全局敏感性分析与贝叶斯标定——不可行，每次求解约需 10^3 s。高斯过程与多项式混沌代理在非线性耦合设置下受维度限制；物理信息神经网络需要闭合残差形式，而迭代求解器交换无法提供；确定性神经网络仅产生点估计。上述方案均无法提供三类分析共同所需的连贯预测分布。

A Bayesian neural network (BNN) fills this gap. By placing a distribution over the network weights, a BNN yields a posterior predictive distribution that can be sampled at negligible cost once trained. This

single distribution serves as the shared probabilistic layer on which forward propagation, variance-based sensitivity attribution and observation-conditioned posterior calibration all operate, with no retraining or recalibration between them.

贝叶斯神经网络（BNN）填补了这一空缺。通过对权重引入分布，BNN 在训练完成后即可以极低代价反复采样其后验预测分布。该分布充当前向传播、基于方差的敏感性归因与观测条件后验标定共同使用的概率层，三者之间无需重训或重新校准。

This paper presents a BNN-based posterior-predictive framework for end-to-end probabilistic analysis of coupled multi-physics systems. The framework unifies forward uncertainty propagation, Sobol sensitivity decomposition and MCMC posterior calibration on a single predictive layer trained on coupled OpenMC–FEniCS data of a heat-pipe-cooled reactor. It reveals that two-way coupling damps stress variability by approximately 28%, that stress and reactivity are governed by largely separate parameter pathways, and that the posterior calibration contracts the material-parameter distribution toward observation-compatible regions—all at five orders of magnitude less cost per evaluation than a coupled high-fidelity solve.

本文提出了一套基于 BNN 后验预测分布的耦合多物理系统端到端概率分析框架。该框架将前向不确定性传播、Sobol 敏感性分解与 MCMC 后验标定统一到基于热管冷却反应堆耦合 OpenMC–FEniCS 数据训练的单一预测层上。框架揭示：双向耦合使应力变异降低约 28%；应力与反应性分别受制于基本独立的参数通道；后验标定将材料参数分布收敛至与观测相容的区域——以上分析的单次评估代价较耦合高保真求解低约五个数量级。

2 Results / 结果

2.1 Surrogate accuracy and calibration / 代理精度与校准

The framework rests on a single BNN that predicts all 15 coupled and decoupled outputs simultaneously; if this surrogate is inaccurate, the downstream propagation, sensitivity and calibration results are unreliable. We evaluated the physics-regularized BNN alongside a reference (unconstrained) BNN and two conventional baselines—MC-Dropout and a five-member deep ensemble—on 435 held-out test samples (Table 1, Fig. 1). Results for the physics-regularized BNN are reported as mean $\pm$ s.d. across 5 random seeds; ablation across four BNN variants is in Supplementary Table S3.

该框架依赖一个同时预测全部 15 个耦合与解耦输出的 BNN；若代理精度不足，下游的传播、敏感性与标定结论均不可靠。我们将物理正则化 BNN 与参考（无约束）BNN 及两种传统基线方法——MC-Dropout 和五成员深度集成——在 435 个留出测试样本上进行了评估（表 1，图 1）。物理正则化 BNN 的结果以 5 个随机种子的均值 $\pm$ 标准差报告；四种 BNN 变体的消融实验见补充表 S3。

On coupled peak global stress the physics-regularized BNN achieves $R^2 = 0.935 \pm 0.007$ (RMSE 7.63 ± 0.47 MPa), comparable to MC-Dropout ($R^2 = 0.934$) and deep ensemble ($R^2 = 0.934$). The BNN advantage is most pronounced on temperature outputs: for maximum fuel temperature the BNN RMSE is 4.26 ± 0.18 K versus 4.50 K for both baselines, and the expected calibration error (ECE) is 0.027 versus 0.183 (MC-Dropout) and 0.190 (deep ensemble)—a five- to six-fold improvement in calibration quality (Fig. 1A, C).

在耦合峰值全局应力上，物理正则化 BNN 达到 $R^2 = 0.935 \pm 0.007$（RMSE 7.63 ± 0.47 MPa），与

MC-Dropout（$R^2 = 0.934$）和深度集成（$R^2 = 0.934$）相当。BNN 的优势在温度输出上最为显著：最大燃料温度的 BNN RMSE 为 4.26 ± 0.18 K，而两种基线方法均为 4.50 K；期望校准误差（ECE）为 0.027，而 MC-Dropout 为 0.183、深度集成为 0.190——校准质量提升五至六倍（图 1A、C）。

The 90% predictive intervals are conservatively calibrated across all outputs (PICP 0.937–1.000), and incorporating physics-prior monotonicity constraints tightens the intervals relative to the unconstrained reference without reducing coverage (Supplementary Table S3). The continuous ranked probability score (CRPS) is lowest for the physics-regularized BNN on all five primary outputs (Table 1).

所有输出的 90% 预测区间均呈保守校准特性（PICP 0.937–1.000），引入物理先验单调性约束可在不降低覆盖率的前提下收窄区间（补充表 S3）。在五个主要输出上，物理正则化 BNN 的连续排序概率评分（CRPS）均为最低（表 1）。

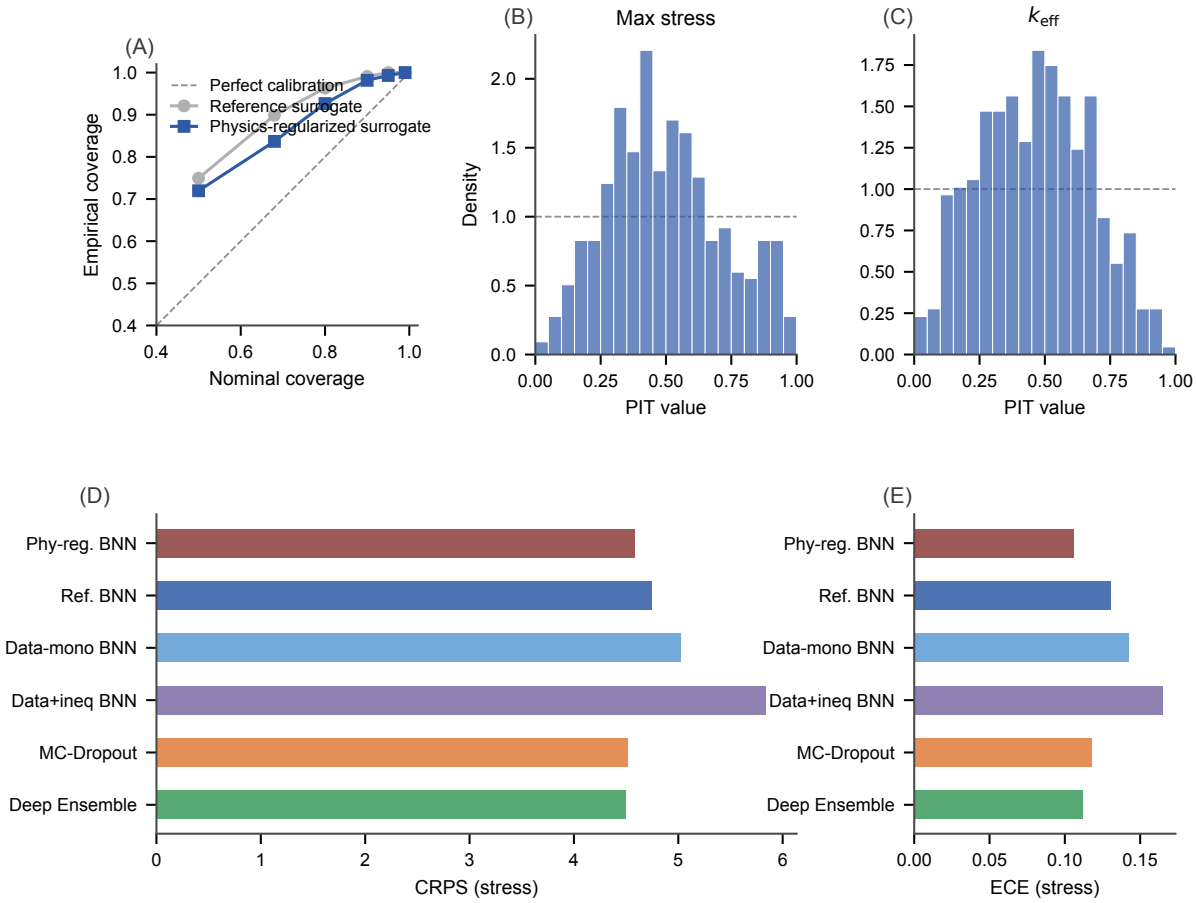

Figure 1: **Surrogate accuracy and uncertainty calibration.** (**A**) Predicted vs. true values for five primary coupled outputs (physics-regularized BNN, 435 test samples). (**B**) Residual distributions. (**C**) Calibration curves comparing BNN, MC-Dropout and deep ensemble. The BNN achieves five- to six-fold lower ECE than the baselines on temperature outputs.

2.2 Physics consistency / 物理一致性

We verified the BNN's adherence to known physical relationships through two tests: monotonicity verification against 15 input–output pairs derived from the constitutive laws, and inequality constraint satisfaction (Fig. 2A). For all high-confidence monotonicity pairs—including the dominant stress drivers $E_{\text{intercept}}$ (+), α_{base} (+), and k_{ref} (−)—all four BNN variants achieved zero violation rates on the 435 held-out test

Table 1: **Surrogate accuracy on five primary coupled outputs.** Physics-regularized BNN: mean ± s.d. across 5 seeds; external baselines: single run. Bold CRPS indicates best per output.
表 1 | 五个主要耦合输出的代理精度。物理正则化 BNN：5 个种子的均值 ± 标准差；外部基线：单次运行。粗体 CRPS：各输出最优。

Output	R^2	RMSE	PICP	ECE	CRPS
Physics-regularized BNN					
k_{eff}	0.633 ± 0.258	0.0006 ± 0.0004	0.981 ± 0.007	0.123	**0.0002**
Max. fuel temp.	0.621 ± 0.044	4.26 ± 0.18 K	0.936 ± 0.015	0.027	**2.39**
Max. monolith temp.	0.622 ± 0.044	4.80 ± 0.20 K	0.942 ± 0.013	0.031	**2.70**
Max. global stress	0.935 ± 0.007	7.63 ± 0.47 MPa	0.987 ± 0.007	0.106	**4.58**
Wall expansion	0.993 ± 0.001	0.0020 ± 0.0002	1.000 ± 0.000	0.194	**0.0017**
MC-Dropout					
Max. fuel temp.	0.590	4.50 K	—	0.183	2.62
Max. global stress	0.934	7.66 MPa	—	0.118	4.52
Deep Ensemble (5 members)					
Max. fuel temp.	0.590	4.49 K	—	0.190	2.63
Max. global stress	0.934	7.64 MPa	—	0.112	4.50

samples using discrete $\pm\delta$ perturbation testing. The only non-trivial violations occurred for the medium-confidence pair SS316_$\alpha \rightarrow$ fuel temperature (36–54% across models), reflecting a weaker and potentially nonlinear relationship that the training data does not consistently support. All four models also satisfied ordering constraints (max $\geq$ average temperature) and non-negativity constraints (stress ≥ 0) with zero violations.

我们通过两项测试验证了 BNN 对已知物理关系的遵守情况：针对由本构方程推导的 15 个输入–输出对进行单调性验证，以及不等式约束满足性测试（图 2A）。对于所有高置信度单调对——包括应力的主要驱动因素 $E_{\text{intercept}}$（+）、α_{base}（+）和 k_{ref}（−）——四种 BNN 变体在 435 个测试样本上通过离散 $\pm\delta$ 扰动测试均实现了零违反率。唯一的显著违反出现在中等置信度的 SS316_α $\rightarrow$ 燃料温度对（各模型 36–54%），反映了一种较弱且可能非线性的物理关系。所有四个模型在排序约束（最大 $\geq$ 平均温度）和非负约束（应力 ≥ 0）上均实现零违反。

Decomposing total predictive variance into epistemic (weight-uncertainty) and aleatoric (observation-noise) components revealed that aleatoric uncertainty dominates across all outputs, accounting for 58–90% of total variance (Fig. 2B). The epistemic fraction was highest for wall expansion ($\sim$42%) and stress ($\sim$30%), indicating greater model-form uncertainty for these mechanically driven outputs.

将总预测方差分解为认知（权重不确定性）和偶然（观测噪声）分量后发现，偶然不确定性在所有输出上占主导地位，占总方差的 58–90%（图 2B）。认知不确定性占比最高的是壁面膨胀（$\sim$42%）和应力（$\sim$30%），表明模型对这些力学驱动输出的结构不确定性较大。物理正则化略微降低了温度预测的认知不确定性（认知占比 10.1%，而参考 BNN 为 11.2–12.0%），这与单调性约束限制了可信模型结构空间的预期一致。

2.3 Forward uncertainty propagation / 前向不确定性传播

Propagating 20,000 Latin hypercube samples through the BNN posterior predictive mean yields a coupled stress distribution with mean 162.5 MPa and standard deviation 35.5 MPa, versus a decoupled mean of 202.4 MPa and standard deviation of 49.2 MPa. Coupling reduces the stress standard deviation by approx-

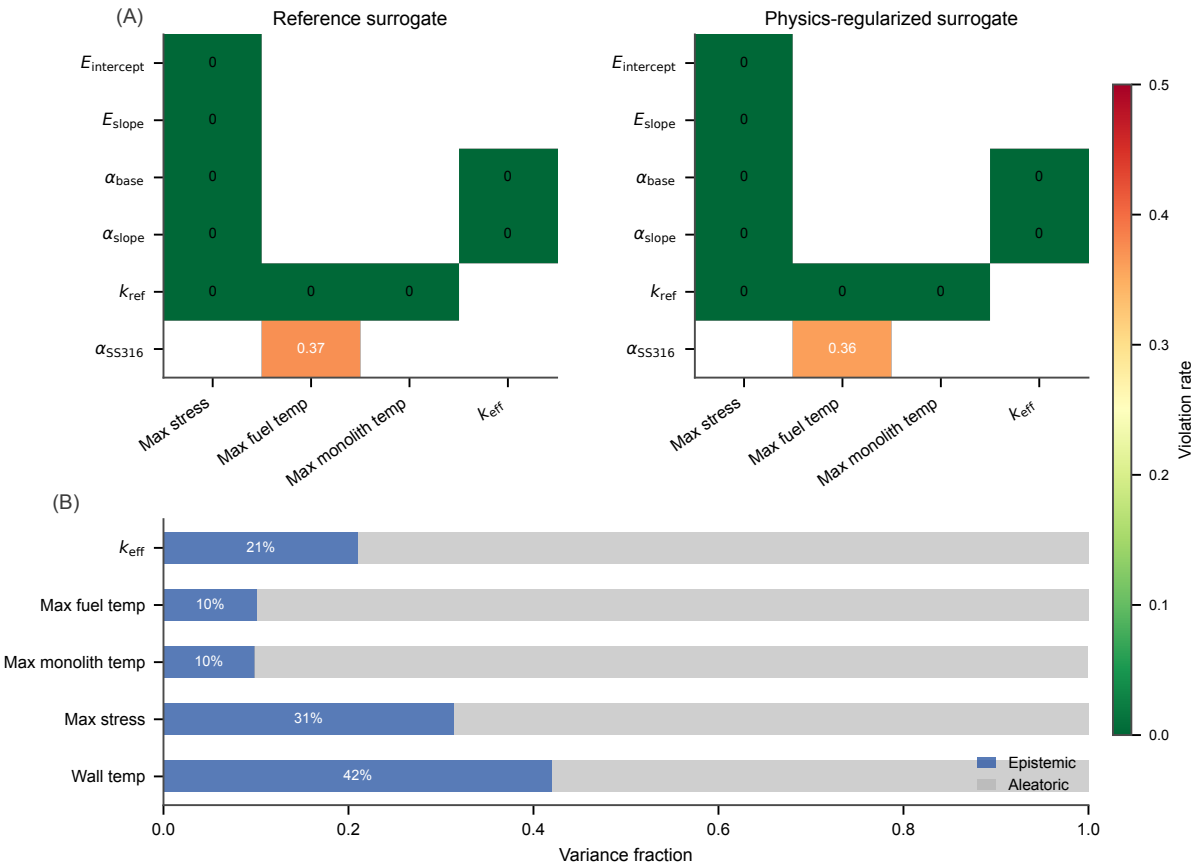

Figure 2: **Physics consistency verification.** (**A**) Monotonicity violation rates for 15 input–output pairs across four BNN variants; high-confidence pairs achieve zero violations. (**B**) Epistemic vs. aleatoric variance decomposition across five primary outputs.

imately 28% and lowers the mean by approximately 40 MPa (Table 2, Fig. 3).

将 20,000 个拉丁超立方采样点通过 BNN 后验预测均值传播，得到耦合应力分布均值 162.5 MPa、标准差 35.5 MPa，而解耦分布均值为 202.4 MPa、标准差 49.2 MPa。耦合使应力标准差降低约 28%，均值降低约 40 MPa （表 2，图 3）。

The coupled k_{eff} has mean 1.1036 and standard deviation approximately 107 pcm. The decoupled k_{eff} is effectively constant because, at the reference geometry, the sampled material parameters do not enter the neutronic cross-section tables; all coupled k_{eff} variability originates from thermal-expansion modification of core geometry.

耦合 k_{eff} 均值为 1.1036，标准差约 107 pcm。解耦 k_{eff} 几乎恒定，因为在参考几何下，采样的材料参数不进入中子截面表；耦合 k_{eff} 的全部变异源自热膨胀对堆芯几何的修改。

Table 2: **Forward UQ summary: coupled vs. decoupled distributions** (20,000 LHS samples through BNN posterior predictive mean).
表 2 | 前向 UQ 汇总：耦合与解耦分布。

Output	Coupled mean	Coupled s.d.	Decoupled mean	Decoupled s.d.
Stress (MPa)	162.5	35.5	202.4	49.2
k_{eff}	1.1036	0.00107	∼constant	∼0

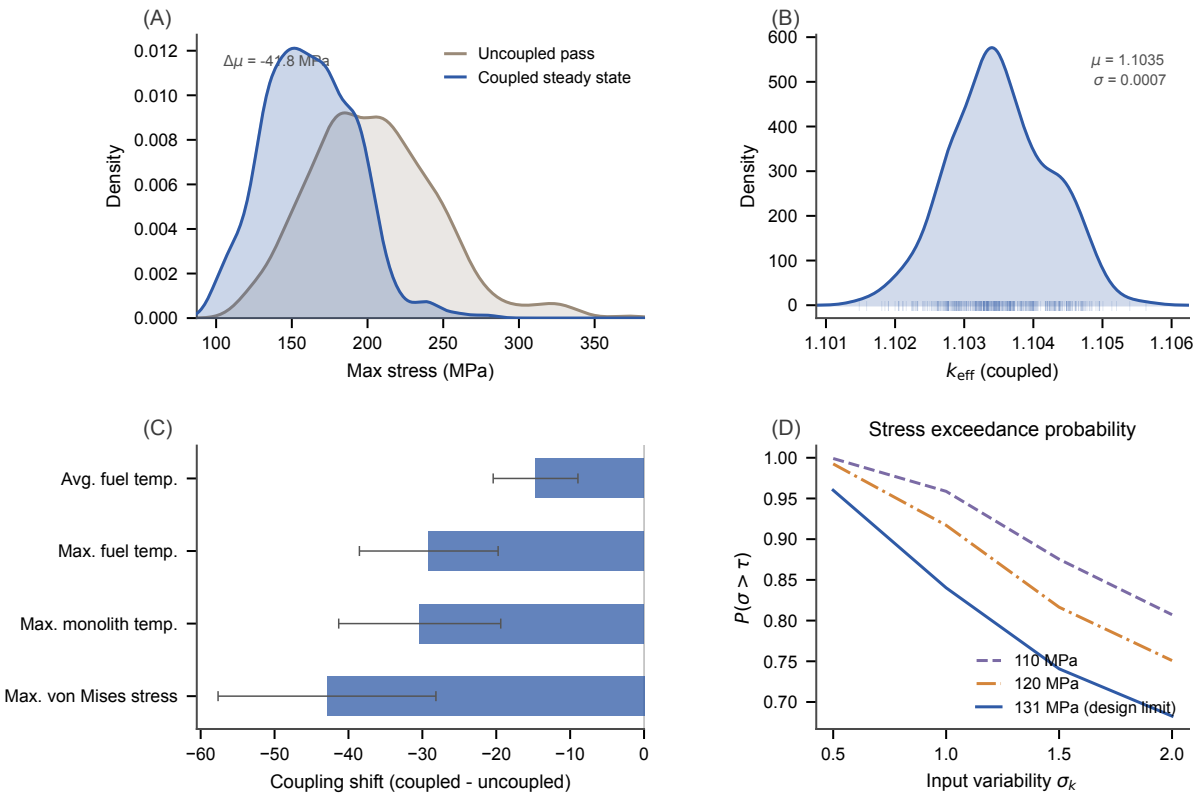

Figure 3: **Forward uncertainty propagation.** Coupled vs. decoupled output distributions for stress and k_{eff}. Two-way coupling reduces the stress standard deviation by ≈28% and introduces k_{eff} variability that is absent in the decoupled case.

2.4 Global sensitivity analysis / 全局敏感性分析

Sobol analysis (50 independent replications, $N_S = 4096$ per matrix; full index table in Supplementary Table S4) shows that stress and k_{eff} are governed by two largely separate physical pathways (Table 3, Fig. 4).

Sobol 分析（50 次独立重复，每矩阵 $N_S = 4096$；完整指标表见补充表 S4）表明应力与 k_{eff} 受制于两条基本独立的物理路径（表 3，图 4）。

Stress is rooted in the elastic-constitutive pathway. $E_{\text{intercept}}$ alone accounts for approximately 55% of the first-order variance ($S_1 = 0.548$, 90% CI [0.498, 0.572]), and the near-zero gap between its total-order and first-order indices indicates that it acts with minimal interaction. Secondary contributors—α_{base} ($S_1 = 0.210$), k_{ref} and ν—add a collective approximately 15%, while the remaining parameters are negligible. The dominance of $E_{\text{intercept}}$ is a direct consequence of the generalised Hooke's law: stiffness enters the stress field linearly.

应力根植于弹性本构路径。$E_{\text{intercept}}$ 单独贡献约 55% 的一阶方差（$S_1 = 0.548$，90% CI [0.498, 0.572]），其全阶与一阶指标之间的差距接近零，表明其作用几乎无交互。次要贡献因素——α_{base}（$S_1 = 0.210$）、k_{ref} 和 ν——合计约贡献 15%，其余参数可忽略。$E_{\text{intercept}}$ 的主导地位是广义胡克定律的直接结果：刚度线性进入应力场。

k_{eff} is rooted in the thermal-expansion pathway. α_{base} accounts for approximately 77% of the first-order variance ($S_1 = 0.771$, 90% CI [0.747, 0.788]): it sets the magnitude of geometric expansion, which alters neutron leakage and shifts the neutronic balance. α_{slope} contributes a further approximately 19% ($S_1 = 0.195$), while no elasticity or conductivity parameter has a first-order effect on reactivity above the noise floor.

k_{eff} 根植于热膨胀路径。α_{base} 贡献约 77% 的一阶方差（$S_1 = 0.771$，90% CI [0.747, 0.788]）：它设定了几何膨胀的幅度，进而改变中子泄漏并移动中子学平衡点。α_{slope} 进一步贡献约 19%（$S_1 = 0.195$），而弹性或热导率参数对反应性的一阶效应均在噪声底以下。

The two pathways share almost no dominant parameters. This separation means that stress and k_{eff} observations constrain different regions of the material-parameter space—an important constraint for any monitoring strategy that aims to reduce material-property uncertainty.

两条路径几乎不共享主导参数。该分离意味着应力与 k_{eff} 的观测约束了材料参数空间的不同区域——这对任何旨在降低材料属性不确定性的监测策略而言是一个重要约束。

Convergence analysis confirmed that these rankings stabilised by $N_S = 2048$, with standard deviations below 0.04 at the final sample size of 8192 (Supplementary Fig. S5). Both the reference and physics-regularized BNN yielded consistent rankings, indicating that the regularization does not distort the sensitivity structure.

收敛性分析表明，上述排名在 $N_S = 2048$ 时已趋于稳定，在最终样本量 8192 处标准差低于 0.04（补充图 S5）。参考 BNN 与物理正则化 BNN 给出了一致的排名，表明正则化未扭曲敏感性结构。

Table 3: **First-order Sobol indices S_1 for stress and k_{eff}** (physics-regularized BNN, $N_S = 4096$, 50 replications, 90% CI). Only factors with CI lower bound > 0 are shown.
表 3 | 应力与 k_{eff} 的一阶 Sobol 指标 S_1。仅列出 CI 下界 > 0 的因素。

Parameter	Stress S_1 [90% CI]	k_{eff} S_1 [90% CI]
$E_{\text{intercept}}$	0.548 [0.498, 0.572]	—
α_{base}	0.210 [0.171, 0.254]	0.771 [0.747, 0.788]
ν	0.063 [0.034, 0.097]	—
k_{ref}	0.065 [0.014, 0.118]	—
α_{slope}	—	0.195 [0.112, 0.250]

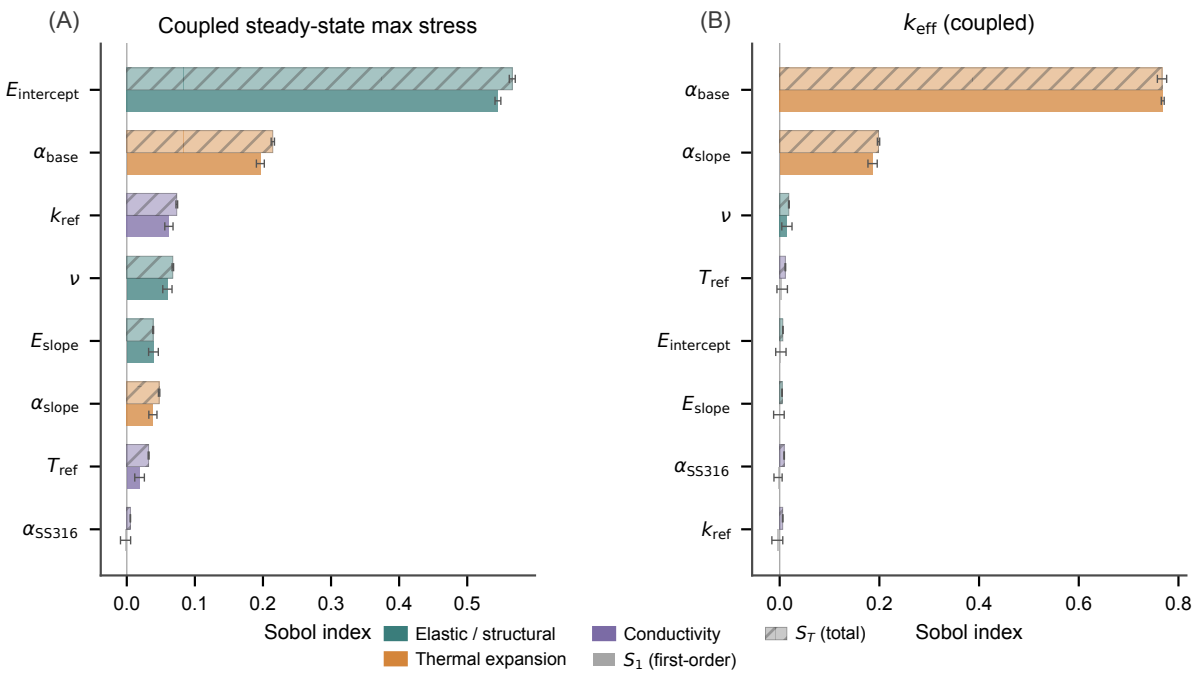

Figure 4: **Sobol sensitivity analysis.** First-order and total-order indices for stress and k_{eff} reveal two largely separate physical pathways: the elastic-constitutive pathway (dominated by $E_{\text{intercept}}$) for stress, and the thermal-expansion pathway (dominated by α_{base}) for k_{eff}.

2.5 Posterior calibration / 后验标定

Bayesian calibration uses the same BNN to update the prior parameter distribution given noisy observations. The four calibrated parameters—$E_{\text{intercept}}$, α_{base}, α_{slope} and k_{ref}—are the four that the Sobol analysis identified as covering the dominant variance channels for stress and k_{eff}. Metropolis–Hastings MCMC is run on 18 benchmark cases drawn from the test split (6 low / 6 near / 6 high stress; details in Methods) with 4 independent chains per case.

贝叶斯标定利用同一 BNN 在给定含噪观测数据的条件下更新先验参数分布。四个标定参数——$E_{\text{intercept}}$、α_{base}、α_{slope} 和 k_{ref}——是 Sobol 分析所识别的覆盖应力与 k_{eff} 主要方差通道的因素。Metropolis–Hastings MCMC 在 18 个基准算例（取自测试集：低应力 6 例 / 阈值附近 6 例 / 高应力 6 例，详见方法部分）上运行，每个案例使用 4 条独立链。

Chain convergence was assessed using rank-normalised split-$\hat{R}$ (Vehtari et al. 2021) and Geyer initial-positive-sequence effective sample size. Across all 72 parameter–case combinations for the physics-regularized BNN, $\hat{R}_{\text{max}} = 1.017$ (well below the 1.1 threshold), $\text{ESS}_{\text{min}} = 352$, and mean acceptance rate $= 0.605$ (Fig. 5A–B; Supplementary Table S5).

链收敛性通过秩归一化 split-$\hat{R}$（Vehtari 等，2021）和 Geyer 初始正序列有效样本量进行评估。在物理正则化 BNN 的全部 72 个参数–案例组合中，$\hat{R}_{\text{max}} = 1.017$（远低于 1.1 阈值），$\text{ESS}_{\text{min}} = 352$，平均接受率 $= 0.605$（图 5A–B；补充表 S5）。

Across the 18 cases, the posterior 90%-CI coverage is 0.917—the observations successfully constrain the parameters that the Sobol analysis flagged as most influential. In higher-stress cases, the $E_{\text{intercept}}$ marginal shifts toward larger values, consistent with its dominant role in the elastic-constitutive pathway. The joint posterior of $E_{\text{intercept}}$ and α_{base} develops a compensation ridge: an increase in stiffness can be partially offset by a decrease in thermal expansion, leaving the stress product $\sigma \sim E \cdot \alpha \cdot \Delta T$ approximately invariant. This partial non-identifiability arises from the multiplicative structure and would not be visible in a single-output calibration (Fig. 5).

18 例的后验 90%-CI 覆盖率为 0.917——观测数据成功约束了 Sobol 分析标记为最具影响力的参数。在高应力算例中，$E_{\text{intercept}}$ 的边缘分布向更大值偏移，这与其在弹性本构路径中的主导作用一致。$E_{\text{intercept}}$ 与 α_{base} 的联合后验形成补偿脊：刚度的增加可被热膨胀的降低部分抵消，使应力乘积 $\sigma \sim E \cdot \alpha \cdot \Delta T$ 近似不变。这种部分不可辨识性源于乘法结构，在单输出标定中不可见（图 5）。

At the posterior-mean parameters of each case, the BNN-predicted stress tracks the true value with MAE $= 10.4$ MPa over a true-stress range of 102–198 MPa. Prior sensitivity analysis (6 prior configurations) and observation noise sensitivity analysis (noise fraction 0.5–10%) confirmed the robustness of posterior inference (Supplementary Figs. S7–S8, Tables S11–S12).

在各算例的后验均值参数处，BNN 预测应力跟踪真实值的 MAE $= 10.4$ MPa，真实应力范围为 102–198 MPa。先验敏感性分析（6 种先验配置）和观测噪声敏感性分析（噪声分数 0.5–10%）确认了后验推断的鲁棒性（补充图 S7–S8，补充表 S11–S12）。

2.6 Computational efficiency / 计算效率

The BNN produces one posterior-predictive draw (averaged over 50 Monte Carlo weight samples) in 1.29×10^{-2} s on GPU. Batched inference at batch size 1024 reaches a per-sample cost of 1.34×10^{-5} s, equivalent

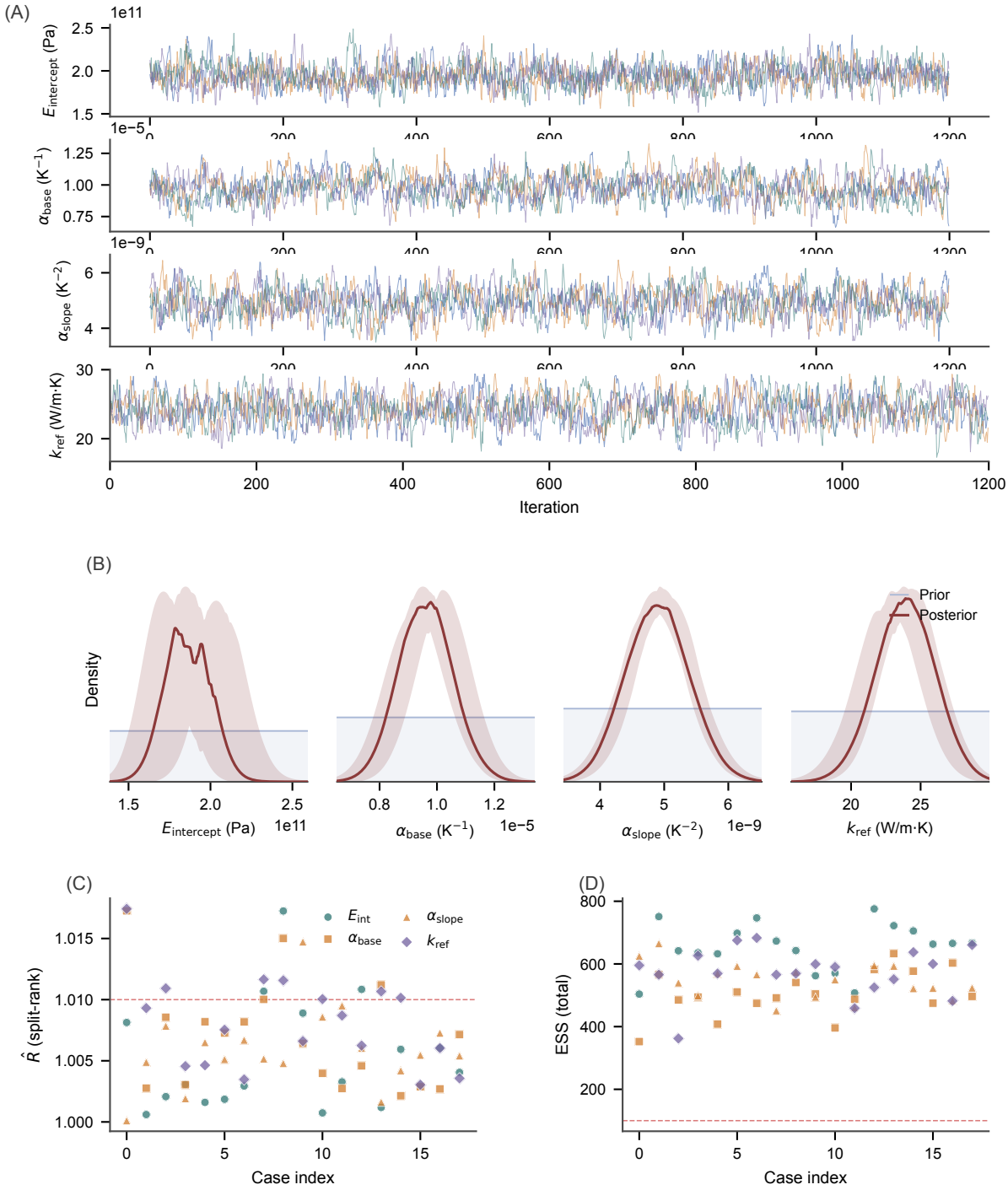

Figure 5: **Posterior calibration results.** (**A**) MCMC convergence diagnostics (split-$\hat{R}$ and ESS). (**B**) Acceptance rates across 18 benchmark cases. (**C**) Marginal posterior distributions for $E_{\text{intercept}}$ and α_{base} in representative low-, near-, and high-stress cases. (**D**) Joint posterior of $E_{\text{intercept}}$ vs. α_{base} showing the compensation ridge.

to a throughput of approximately 7.5×10^4 samples/s. The coupled HF solver requires 2266 s (mean of three independent runs, s.d. 1.34 s; AMD EPYC 9654 96-core node, 566 GB RAM, 2× RTX 5090). The resulting speedup is approximately 1.76×10^5 for one posterior-predictive draw and approximately 1.69×10^8 for batched per-sample inference (Fig. 6A). A 20,000-sample forward-UQ pass requires approximately 0.3 s of GPU time, against 4.5×10^7 s of sequential HF cost. This speedup enables the full probabilistic workflow—20,000-sample Monte Carlo propagation, 50-replication Sobol analysis, and 18-case MCMC calibration—to run within minutes of wall-clock time.

BNN 在 GPU 上完成一次后验预测抽样（50 次 Monte Carlo 权重采样均值化）耗时 1.29×10^{-2} s。批量推理（batch size $= 1024$）的单样本代价为 1.34×10^{-5} s，等效吞吐量约 7.5×10^4 样本/s。耦合高保真求解器耗时 2266 s（三次独立运行均值，标准差 1.34 s；AMD EPYC 9654 96 核节点，566 GB RAM，2× RTX 5090）。由此得到的加速比：单次后验预测抽样约 1.76×10^5 倍，批量单样本推理约 1.69×10^8 倍。20,000 样本的前向 UQ 传播仅需约 0.3 s GPU 时间，而序贯高保真代价为 4.5×10^7 s。该加速使得完整的概率工作流——20,000 样本 Monte Carlo 传播、50 次 Sobol 重复分析和 18 例 MCMC 标定——可在数分钟挂钟时间内完成。

The BNN also exploits limited training data efficiently. With only 507 training samples (25% of the full training set), the surrogate achieves RMSE $= 4.93$—96% of its peak performance at full data utilisation (RMSE $= 4.62$). Marginal improvement from 50% to 100% data is less than 1% (Fig. 6B).

BNN 对有限训练数据的利用效率也很高。仅需 507 个训练样本（完整训练集的 25%），代理即可达到 RMSE $= 4.93$——峰值性能（RMSE $= 4.62$）的 96%。从 50% 到 100% 数据的边际改善不到 1%（图 6B）。

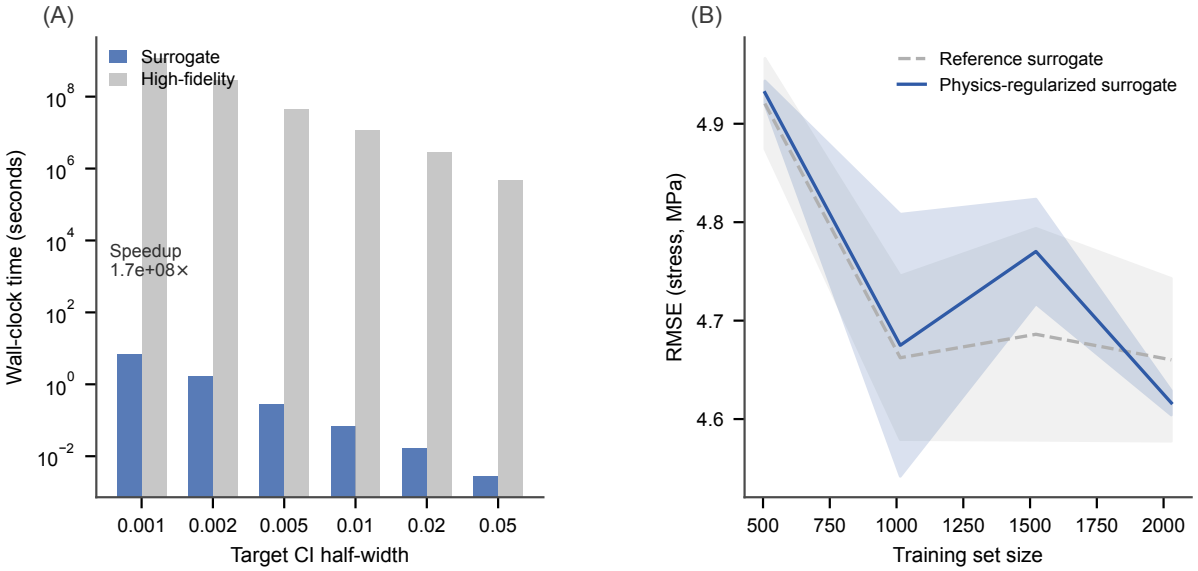

Figure 6: **Computational efficiency. / 计算效率。**(**A**) Speedup of BNN surrogate relative to coupled HF solver: single posterior-predictive draw ($1.76 \times 10^5\times$) and batched per-sample inference ($1.69 \times 10^8\times$). BNN 代理相对于耦合 HF 求解器的加速比：单次后验预测抽样（1.76×10^5 倍）和批量单样本推理（1.69×10^8 倍）。(**B**) Data efficiency: test RMSE as a function of training set size, showing near-saturated performance at 25% data utilisation. 数据效率：测试 RMSE 随训练集大小的变化，25% 数据利用率下性能已接近饱和。

3 Discussion / 讨论

The stress standard deviation drops from approximately 49.2 MPa in the decoupled response to approximately 35.5 MPa after coupling convergence, a roughly 28% reduction. The mechanism is the negative neutronic–thermal feedback characteristic of compact monolithic-core systems: thermal expansion modifies the core geometry, which shifts the neutronic balance and partially offsets the direct stress increase. What the BNN-based forward propagation establishes is the net effect of this feedback on the output distribution. A decoupled analysis does not see the feedback loop and therefore overestimates the parameter-driven stress variability. The overestimate is conservative for safety-margin evaluation and cannot produce an unsafe conclusion, but it inflates the apparent penalty of material-property uncertainty. In practice, this can translate into manufacturing tolerances stricter than the coupled response warrants.

应力标准差从解耦响应的约 49.2 MPa 降至耦合收敛后的约 35.5 MPa，降幅约 28%。其机制是紧凑型一体式堆芯系统特有的负中子–热反馈：热膨胀修改堆芯几何，移动中子学平衡点，从而部分抵消直接的应力增加。基于 BNN 的前向传播所建立的是该反馈对输出分布的净效应。解耦分析无法感知反馈回路，因而高估了参数驱动的应力变异。这种高估对安全裕度评估而言是保守的，不会导致不安全的结论，但会放大材料属性不确定性的表观代价。在工程实践中，这可能转化为比耦合响应所需更严格的制造公差。

Compared with an ELBO-only BNN (Supplementary Table S3), adding physics-prior monotonicity constraints shrinks the 90% predictive interval without reducing coverage. The primary model enforces monotonicity relations derived from the physics of the constitutive laws—for example, that stress increases with Young's modulus at fixed thermal strain. The interval tightening at constant coverage indicates genuine sharpening rather than miscalibration, and the same posterior predictive distribution propagates consistently through forward UQ, Sobol attribution and MCMC calibration. A point-estimate surrogate would require separate uncertainty wrappers for each analysis with no guarantee of mutual consistency.

与仅 ELBO 的 BNN（补充表 S3）相比，引入物理先验单调性约束可在不降低覆盖率的前提下收窄 90% 预测区间。主模型强制执行从本构律物理中推导的单调性关系——例如，在固定热应变下应力随杨氏模量递增。覆盖率不变条件下区间收窄表明这是真实的锐化而非误校准，且同一后验预测分布在前向 UQ、Sobol 归因和 MCMC 标定中保持一致传播。点估计代理需要为每类分析单独构建不确定性包装器，且无法保证彼此一致。

Out-of-distribution analysis confirms that the BNN's epistemic uncertainty inflates appropriately in under-represented input regions: epistemic standard deviation increases by 7–21% in peripheral-tail subsets, while 90% coverage remains above 96% even in these regions (Supplementary Fig. S4). This Bayesian self-awareness is a structural advantage over frequentist baselines that lack a mechanism for signalling elevated model uncertainty.

分布外分析表明，BNN 的认知不确定性在训练覆盖不足的输入区域能够适当膨胀：在外围尾部子集中，认知标准差增加 7–21%，而 90% 覆盖率在这些区域仍保持在 96% 以上（补充图 S4）。这种贝叶斯自感知能力是相对于频率派基线方法的结构性优势——后者缺乏标示模型不确定性升高的机制。

The sensitivity and calibration results have practical consequences for the engineering workflow. For stress-margin assessment, tightening the uncertainty on $E_{\text{intercept}}$—e.g. through coupon tensile tests at operating temperature—is the single highest-leverage action; for reactivity stability, dilatometry to refine

α_{base} is the priority. Because these two measurements target different experimental setups, they can proceed in parallel. When measured data become available, the posterior calibration folds that information into the same surrogate without retraining. If a posterior update reveals that a safety margin is tighter than initially expected, the Sobol ranking identifies which parameter to prioritise in the next round of characterisation. The near-zero parameter overlap between the stress and k_{eff} pathways means that a calibration protocol combining both outputs provides stronger parameter identification than either alone. These indications are model-driven and require verification on the high-fidelity solver before adoption as design directives.

敏感性与标定结果对工程工作流具有实际意义。对于应力裕度评估，收紧 $E_{\text{intercept}}$ 的不确定性——例如通过运行温度下的试样拉伸试验——是单一最高杠杆率操作；对于反应性稳定性，通过膨胀仪细化 α_{base} 是优先事项。由于这两类测量针对不同的实验装置，可以并行推进。当实测数据可用时，后验标定可将该信息纳入同一代理而无需重训。若后验更新表明安全裕度比初始预期更紧，Sobol 排名可识别下一轮表征中应优先的参数。应力与 k_{eff} 路径之间几乎零参数重叠意味着组合两种输出的标定协议可提供比单独任一更强的参数辨识能力。这些指示是模型驱动的，在作为设计指令采纳之前需在高保真求解器上验证。

The surrogate is trained on data from a single HPR design point; a different geometry, fuel enrichment or operating envelope would require regenerating the training set and retraining. The methodology itself is not design-specific: the same workflow of coupled-data generation, BNN training and three-stage probabilistic analysis is directly applicable to other reactor types and other coupled multi-physics systems. The MCMC calibration fixes four parameters at their true values and calibrates the remaining four; extending to a full eight-parameter calibration would require additional independent observations. The k_{eff} surrogate degrades on out-of-distribution α_{base} tails (R^2 drops to approximately 0.29 versus 0.825 in-distribution; Supplementary Fig. S4), so the Sobol and posterior conclusions for k_{eff} hold only within the training range. All posterior summaries are computed on the surrogate side. An independent high-fidelity re-evaluation at the posterior-mean parameter vectors is documented separately. The present study uses synthetic observations constructed from the high-fidelity solver; if experimental or monitoring data become available, the same calibration workflow can ingest real observations directly.

代理在单一 HPR 设计点的数据上训练；不同的几何、燃料富集度或运行包络需要重新生成训练集并重训。方法学本身不依赖于特定设计：同样的耦合数据生成、BNN 训练和三阶段概率分析工作流可直接适用于其他堆型及其他耦合多物理系统。MCMC 标定将四个参数固定为真实值并标定其余四个；扩展到完整的八参数标定需要额外的独立观测。k_{eff} 代理在分布外 α_{base} 尾部性能下降（R^2 从分布内的 0.825 降至约 0.29；补充图 S4），因此 k_{eff} 的 Sobol 和后验结论仅在训练范围内成立。所有后验汇总均在代理端计算。在后验均值参数向量处的独立高保真再评估已另行记录。本研究使用高保真求解器构造的合成观测；若实验或监测数据可用，同一标定工作流可直接纳入真实观测。

4 Methods / 方法

4.1 Material uncertainty model / 材料不确定性模型

The coupled HPR simulation involves eight material properties of the SS316 monolith grouped by physical function: structural mechanics ($E_{\text{intercept}}$, E_{slope}, ν), thermal deformation (α_{base}, α_{slope}), and thermal

transport (k_{ref}, T_{ref}, k_{slope}). All eight are assigned independent uniform priors with half-width 10% of the nominal design value. The same prior distribution is used throughout: as the sampling distribution in forward UQ, as the input domain for Sobol analysis, and as the prior in posterior calibration. Full parameter definitions, nominal values and constitutive-law relationships are in Supplementary Note E and Supplementary Table S1.

耦合 HPR 仿真涉及 SS316 一体化结构的八个材料属性，按物理功能分组：结构力学（$E_{\text{intercept}}$、E_{slope}、ν）、热变形（α_{base}、α_{slope}）和热输运（k_{ref}、T_{ref}、k_{slope}）。八个参数均赋予独立均匀先验，半宽为标称设计值的 10%。同一先验分布贯穿始终：作为前向 UQ 的采样分布、Sobol 分析的输入域以及后验标定的先验。完整参数定义、标称值和本构关系见补充说明 E 和补充表 S1。

4.2 Coupled high-fidelity simulation / 耦合高保真仿真

The high-fidelity workflow couples OpenMC Monte Carlo neutronics with the FEniCS finite-element solver through a Picard fixed-point iteration scheme[9]. Two fidelity levels are retained: a decoupled single-pass prediction before coupling convergence and a coupled steady-state prediction after full convergence. Each HF evaluation produces 15 scalar outputs. A dataset of $n = 2900$ samples was generated by Latin hypercube sampling; samples were split into training (2029), validation (436) and test (435) sets with a fixed random seed (seed $= 2026$). The BNN maps directly from the 8 input parameters to the 15 outputs without resolving intermediate coupling fields. Details of the iteration scheme, convergence criteria and output definitions are in Supplementary Note E.

高保真工作流通过 Picard 不动点迭代格式耦合 OpenMC Monte Carlo 中子学与 FEniCS 有限元求解器[9]。保留两个保真度水平：耦合收敛前的解耦单次预测和完全收敛后的耦合稳态预测。每次高保真评估产生 15 个标量输出。通过拉丁超立方采样生成 $n = 2900$ 的数据集；样本以固定随机种子（seed $= 2026$）分为训练集（2029）、验证集（436）和测试集（435）。BNN 直接从 8 个输入参数映射到 15 个输出，不解析中间耦合场。迭代格式、收敛准则和输出定义详见补充说明 E。

4.3 Bayesian neural network surrogate / 贝叶斯神经网络代理

The surrogate is a BNN with mean-field variational inference (MFVI, Bayes-by-Backprop[10]). The network uses a fully connected backbone with two output heads: a mean head and a log-variance head that models heteroscedastic observation noise. Training minimises the negative evidence lower bound (ELBO). The posterior predictive distribution at a new input is approximated by Monte Carlo averaging over M weight samples from the trained variational posterior: at inference $M = 50$, for Sobol analysis $M = 30$, and for the MCMC likelihood $M = 20$. Hyperparameters were optimised via Optuna (Supplementary Table S2). Physics-prior monotonicity constraints are incorporated as an auxiliary loss term during training. Full ELBO derivation, KL annealing schedule and architecture details are in Supplementary Note A.

代理为采用均场变分推断（MFVI，Bayes-by-Backprop[10]）的 BNN。网络使用全连接骨干，带两个输出头：均值头和建模异方差观测噪声的对数方差头。训练最小化负证据下界（ELBO）。新输入处的后验预测分布通过对训练后变分后验的 M 次权重采样进行 Monte Carlo 平均来近似：推理时 $M = 50$，Sobol 分析时 $M = 30$，MCMC 似然计算时 $M = 20$。超参数通过 Optuna 优化（补充表 S2）。物理先验单调性约束作为辅助损失项在训练中引入。完整 ELBO 推导、KL 退火策略和架构细节见补充说明 A。

4.4 Forward uncertainty propagation / 前向不确定性传播

Forward UQ draws $N = 20{,}000$ Latin hypercube samples from the joint prior and evaluates the BNN posterior predictive mean at each sample. Using the predictive mean rather than a full predictive draw isolates parametric uncertainty from the BNN's own epistemic and aleatoric spread; the predictive-draw variant is reported in Supplementary Note D.

前向 UQ 从联合先验中抽取 $N = 20{,}000$ 个拉丁超立方样本，并在每个样本处评估 BNN 后验预测均值。使用预测均值而非完整预测抽样可将参数不确定性与 BNN 自身的认知和随机展幅隔离；预测抽样变体在补充说明 D 中报告。

4.5 Sobol sensitivity analysis / Sobol 敏感性分析

Global sensitivity is assessed through variance-based Sobol indices estimated with the Saltelli cross-matrix scheme ($N_S = 4096$ per matrix, 40,960 surrogate evaluations per replication). The analysis is repeated 50 times; a factor is treated as a stable contributor only if the lower bound of its 90% CI exceeds zero. Methodological details including the Jansen estimator definition are in Supplementary Note B.

全局敏感性通过基于方差的 Sobol 指标评估，采用 Saltelli 交叉矩阵方案（每矩阵 $N_S = 4096$，每次重复 40,960 次代理评估）。分析重复 50 次；仅当因素的 90% CI 下界超过零时，才将其视为稳定贡献因素。方法细节（包括 Jansen 估计量定义）见补充说明 B。

4.6 Posterior calibration / 后验标定

Posterior calibration uses Bayesian inference: $p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta})$, with the likelihood evaluated through the BNN posterior predictive mean. Four parameters are calibrated ($E_{\text{intercept}}$, α_{base}, α_{slope}, k_{ref}), selected on the basis of their Sobol ranking; the remaining four are fixed at their true values. The 18 benchmark cases are drawn from the test split (6 low, 6 near, 6 high stress). Observations are the true HF outputs of 5 primary coupled quantities perturbed by 2% multiplicative Gaussian noise. The posterior is sampled by Metropolis–Hastings MCMC with 4 independent chains (8000 iterations, burn-in 2000, thinning 5). Implementation details and convergence diagnostics are in Supplementary Note C.

后验标定采用贝叶斯推断：$p(\boldsymbol{\theta} \mid \mathbf{y}_{\text{obs}}) \propto p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta})$，似然通过 BNN 后验预测均值评估。标定四个参数（$E_{\text{intercept}}$、α_{base}、α_{slope}、k_{ref}），基于其 Sobol 排名选取；其余四个参数固定为真实值。18 个基准算例取自测试集（低应力 6 例、阈值附近 6 例、高应力 6 例）。观测为 5 个主要耦合量的真实高保真输出加 2% 乘性高斯噪声。后验通过 Metropolis–Hastings MCMC 采样，含 4 条独立链（8000 次迭代，预热 2000 次，稀疏化步长 5）。实现细节和收敛诊断见补充说明 C。

Data availability / 数据可用性

The coupled simulation dataset and fixed train/validation/test split used in this study are available at [repository URL upon acceptance].

本研究使用的耦合仿真数据集及固定的训练/验证/测试划分可通过 [接受后提供仓库 URL] 获取。

Code availability / 代码可用性

The BNN training, evaluation and analysis code are available at [repository URL upon acceptance].

BNN 训练、评估和分析代码可通过 [接受后提供仓库 URL] 获取。

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Author contributions / 作者贡献

Y.C. developed the BNN surrogate model, designed and performed the experiments, analysed the results and drafted the manuscript. S.W. contributed to the coupled simulation framework. X.L. and T.Z. conceived and supervised the project, provided scientific guidance and revised the manuscript. All authors discussed the results and approved the final version.

陈一诺开发了 BNN 代理模型，设计并执行了实验，分析了结果并撰写了初稿。王思成贡献了耦合仿真框架。刘晓晶和张腾飞构思和指导了本项目，提供了科学指导并修改了论文。所有作者讨论了结果并批准了最终版本。

Competing interests / 利益冲突声明

The authors declare no competing interests.

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